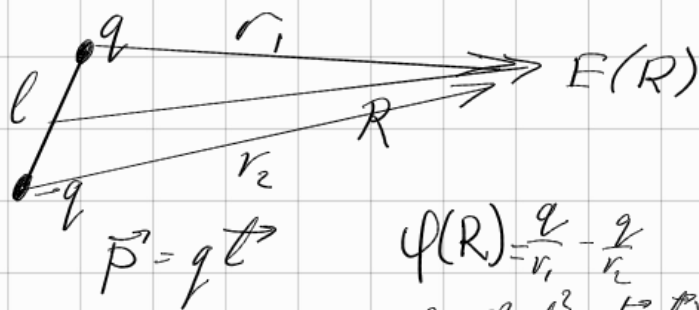




$$\nabla = \vec{i} \frac{\partial}{\partial x} + \vec{j} \frac{\partial}{\partial y} + \vec{k} \frac{\partial}{\partial z}$$

$$\varphi(R) = \frac{q}{r_1} - \frac{q}{r_2}$$

$$r_1^2 = R^2 + \frac{l^2}{4} - 2(\vec{R}; \frac{\vec{l}}{2})$$

$$r_2^2 = R^2 + \frac{l^2}{4} + 2(\vec{R}; \frac{\vec{l}}{2})$$

$$R \gg l, \quad r_1 = R \sqrt{1 + \frac{l^2}{4R^2} - \frac{(\vec{R}; \vec{l})}{R^2}}$$

$$\vec{E} = \frac{q}{r^3} \vec{r}$$

$$\varphi = \frac{q}{r}$$

$$\vec{E} = -\vec{\nabla} \varphi = -q \left(\vec{i} \frac{\partial}{\partial x} + \dots \right)$$

$$r = \sqrt{x^2 + y^2 + z^2}$$

$$\frac{\partial r}{\partial x} = \frac{1}{2} \frac{2x}{r} = \frac{x}{r}$$

$$\vec{\nabla} r = \frac{\vec{r}}{r}$$

$$\varphi = \frac{q}{R} \left[\left(1 + \frac{l^2}{4R^2} - \frac{(\vec{R}; \vec{l})}{R^2} \right)^{-\frac{1}{2}} - \left(1 + \frac{l^2}{4R^2} + \frac{(\vec{R}; \vec{l})}{R^2} \right)^{-\frac{1}{2}} \right] \approx$$

$$\approx \frac{q}{R} \left[1 + \frac{Rl}{2R^2} - 1 + \frac{Rl}{2R^2} \right] = \frac{q(\vec{R}; \vec{l})}{R^3}$$

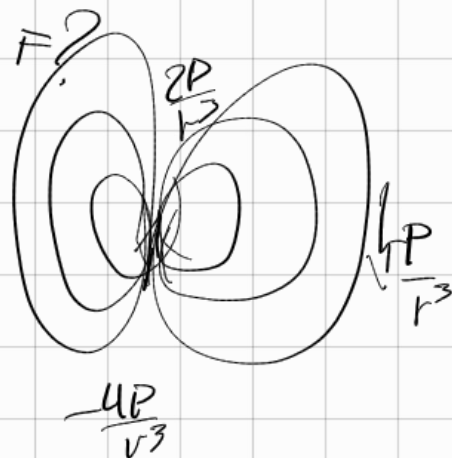
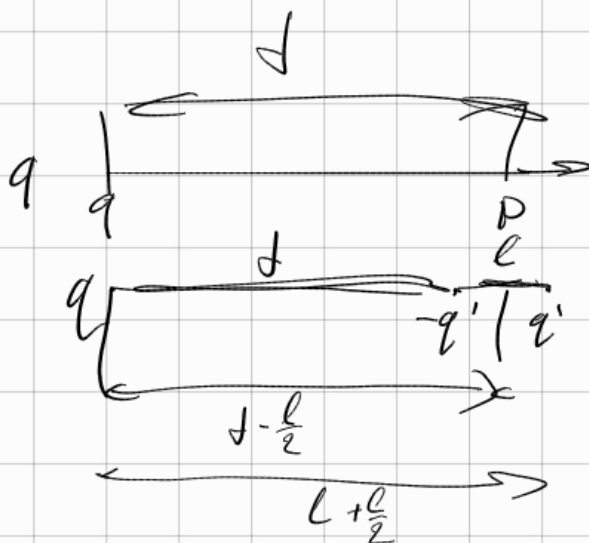
$$-\vec{E} = \vec{\nabla} \varphi = q \vec{\nabla} \frac{(\vec{R}; \vec{l})}{R^3} = q \left[-\frac{3}{R^4} (\vec{R}; \vec{l}) \frac{\vec{R}}{R} + \frac{1}{R^3} \vec{\nabla} (\vec{R}; \vec{l}) \right] \Rightarrow$$

$$\vec{\nabla} (\vec{R}; \vec{l}) = \vec{\nabla} (x l_x + y l_y + z l_z) = l_x \vec{i} \frac{\partial}{\partial x} + l_y \vec{j} \frac{\partial}{\partial y} + l_z \vec{k} \frac{\partial}{\partial z} = \vec{l}$$

$$\Rightarrow -\frac{3(\vec{R}; \vec{l}) \vec{R}}{R^5} + \frac{\vec{l}}{R^3}$$

✓ 1.7

$\vec{l}; \vec{p} \parallel \vec{l}$



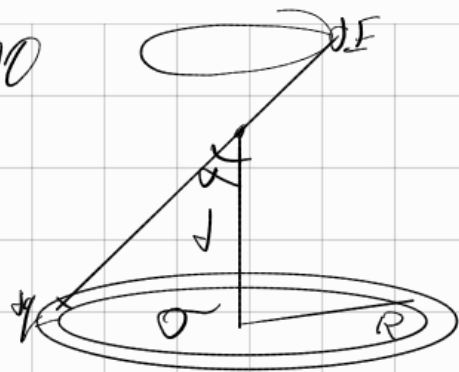
$$E_1 = \frac{q}{(r - \frac{l}{2})^2}$$

$$E_2 = \frac{q}{(r + \frac{l}{2})^2}$$

$$F = -\frac{p}{2} \frac{q}{(r - \frac{l}{2})^2} + \frac{p}{2} \frac{q}{(r + \frac{l}{2})^2} = \frac{pq}{2} \left[\frac{1}{(r - \frac{l}{2})^2} - \frac{1}{(r + \frac{l}{2})^2} \right] \Rightarrow$$

$$\Rightarrow \frac{pq}{2} \left[1 - \frac{l}{r} - \left(1 + \frac{l}{r} \right) \right] = \frac{pq}{2} \left(-2 \frac{l}{r} \right) = -2 \frac{pq}{r^3} = (p; \vec{r}) E$$

1.10



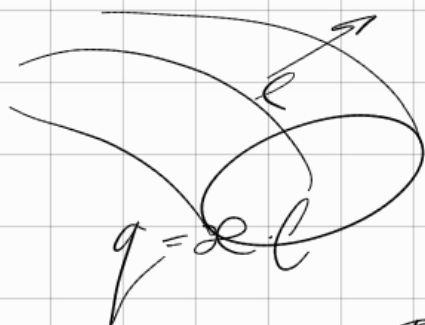
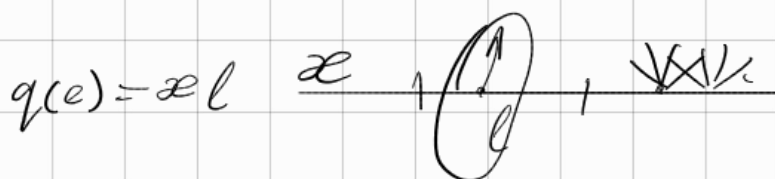
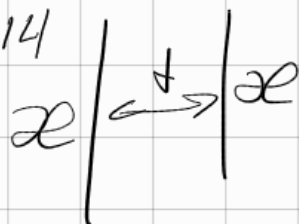
$$\sigma = \frac{Q}{\pi R^2}$$

$$dE = \frac{dq}{d^2 + r^2} = \frac{\sigma 2\pi r dr}{d^2 + r^2}$$

$$E(d) = E_{\text{net}} = \int_0^R 2 \cos \alpha \frac{2\pi r \sigma dr}{d^2 + r^2} = \int_0^R 2 \frac{d 2\pi r \sigma dr}{d^2 + r^2} =$$

$$= 4\pi \sigma d \int_0^R \frac{r dr}{(d^2 + r^2)^{3/2}}$$

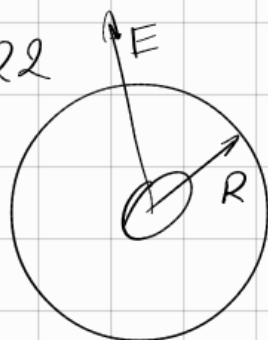
1.14



$$P = 2\pi r \ell E = 4\pi \sigma \ell r^2$$

$$E = \frac{2\sigma \ell}{r}$$

1.22



$$\vec{E} = \frac{q\vec{R}}{R^3}$$



$$E(\rho_+) = \rho_+ \frac{\frac{4}{3}\pi r_i^3}{r^3} \vec{r}_i \quad \left\{ \begin{array}{l} \vec{r}_i \\ \vec{r}_i \end{array} \right.$$

$$= \frac{4}{3}\pi \rho_+ \vec{r}_i$$

$$E(\rho_-) = \rho_- \frac{\frac{4}{3}\pi r_i^3}{r_i^3} \vec{r}_i = \frac{4}{3}\pi \rho_- \vec{r}_i$$

$$\Rightarrow \frac{4}{3}\pi \rho_+ (\vec{r}_i - \vec{r}_i')$$

1.23



$\sigma(\theta)?$



$$E_n \cdot \delta = 4\pi\sigma \delta$$

$$E_r = E_0 \cos\theta \sim \sigma/R$$

$$\sigma(\theta) = C \cdot \cos\theta$$

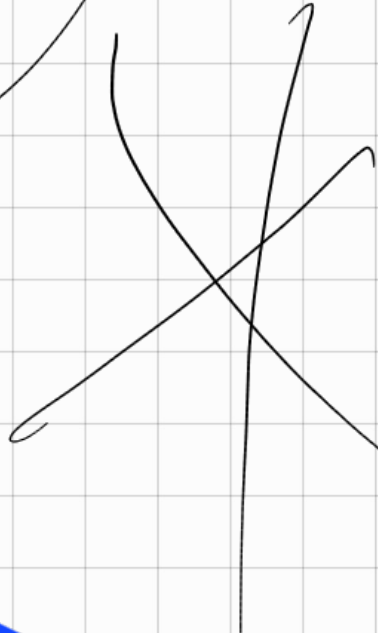
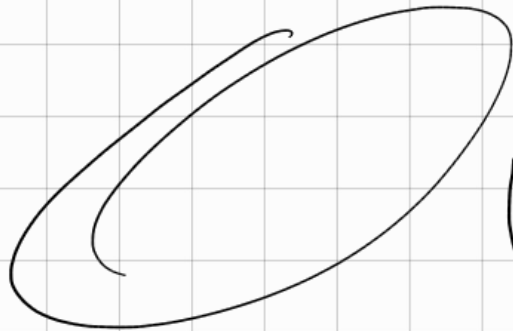
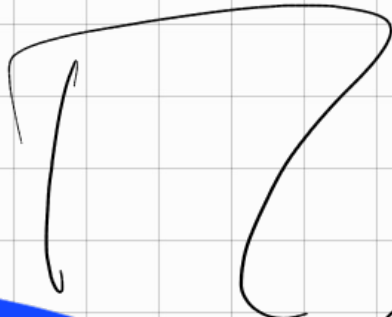
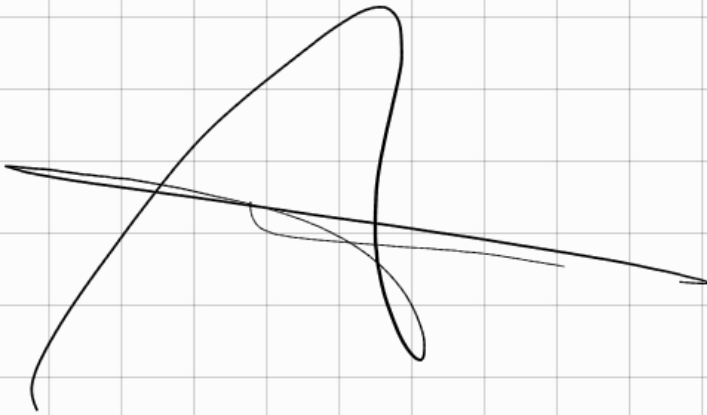
$$E(\theta) = \int_0^\pi \frac{\sigma(\theta) 2\pi R \sin\theta R \cos\theta}{R^2} d\theta$$

$$= 4\pi C \int_0^\pi \cos^2\theta d\cos\theta =$$

$$= \frac{4\pi}{3} (-C) \cos^3\theta \Big|_0^\pi$$

$$= \frac{4}{3}\pi C \cdot 2 = E_0$$

$$\Rightarrow C = \frac{3}{8}\pi E_0$$



КОТО-
ТИЛО



11.09

Homework

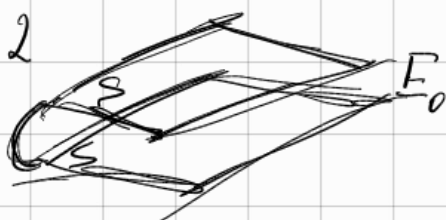
2.1

$$E^2 = m^2 c^4 + p^2 c^2 = (m c^2 + E_k)^2$$

$$P_{ne} = P_{Au}$$

$$U = \frac{q_{ne} q_{Au}}{r_{min}}$$

2.2



$$S = 1.1$$

$$E_0 = 130 \text{ B/m}$$

$$q = ?$$

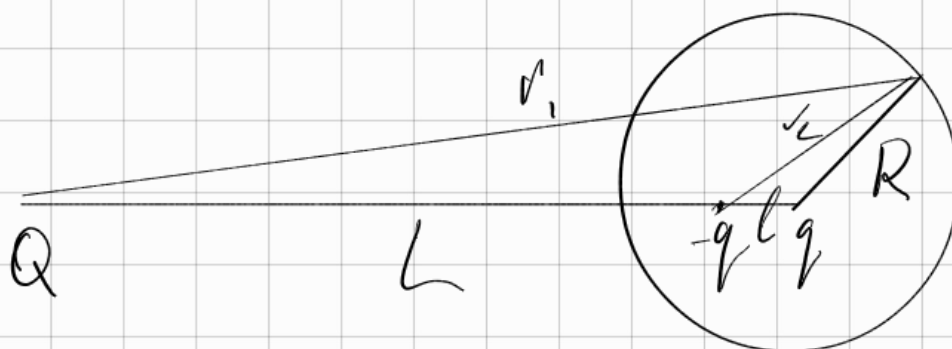
$$\oint \vec{E} \cdot d\vec{S} = 4\pi Q$$



$$ES = 4\pi Q = 4\pi \sigma S_0 = ES_0$$

$$q = \sigma S = \frac{ES}{4\pi}$$

2.3



$$F = \frac{Qq}{\sqrt{L^2 + R^2}} \frac{Q(-q)}{l^2} =$$

$$= \frac{Q^2 R l}{(L^2 + R^2)^2} + \frac{Q^2 R}{l^3}$$

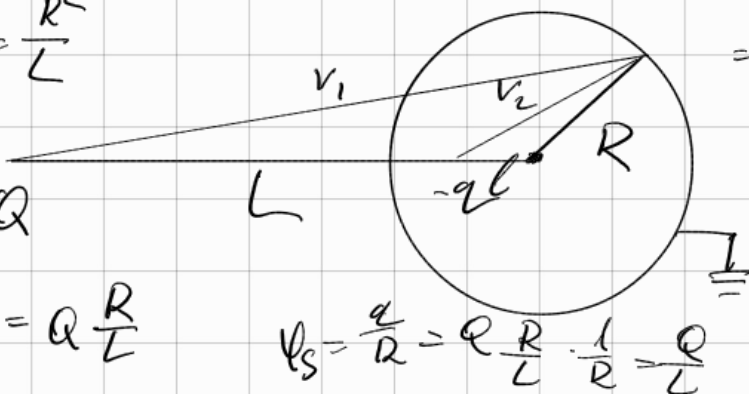
$$\psi_S = 0 \text{ u.s.t.}$$

$$\psi_S = \frac{Q}{r_1} + \frac{q}{r} - \frac{q}{r_2}$$

$$\frac{q}{R} - \frac{r_2}{r_1} = \frac{R}{l} \Rightarrow l = \frac{R^2}{L}$$

$$\psi_S = 0 = \frac{Q}{r_1} = \frac{q}{r_2} \Rightarrow Q$$

$$\Rightarrow q = Q \frac{r_2}{r_1} = Q \frac{R}{L}$$



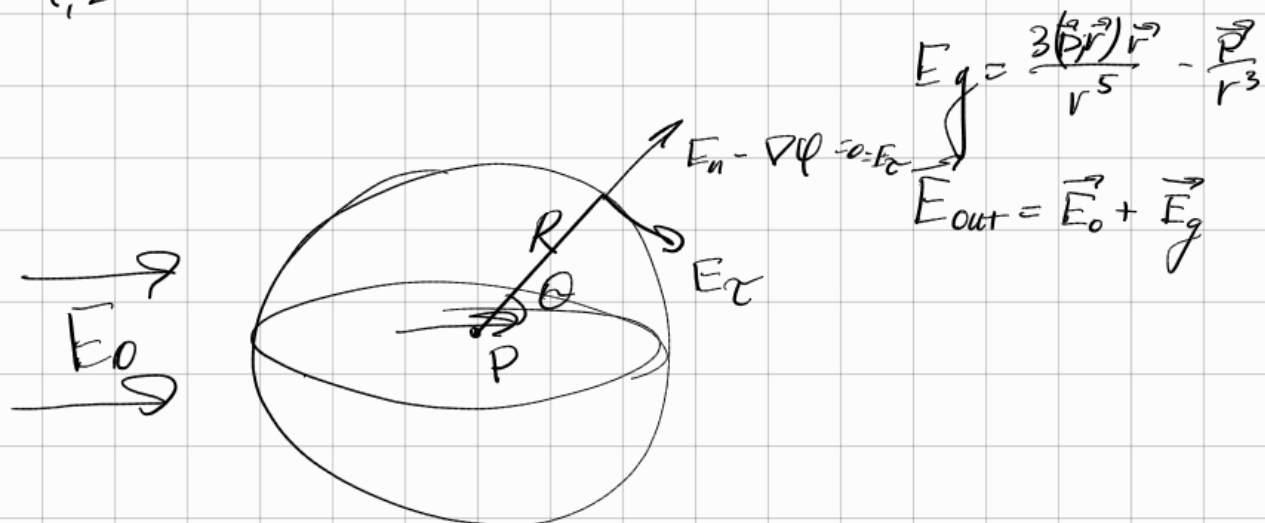
$$F = \frac{Qq}{\sqrt{L^2 + R^2}} = \frac{Q^2 R}{\sqrt{L^2 + R^2}} =$$

$$= \frac{-Q^2 R l}{(L^2 + R^2)^2}$$

$$\psi_S = 0$$

$$\psi_S = \frac{q}{R} = Q \frac{R}{L} \cdot \frac{1}{R} = \frac{Q}{L}$$

1,25



$$E_{out} = E_0 \cos\theta + \frac{3P}{R^3} \cos\theta - \frac{P \cos\theta}{R^3} = \cos\theta \left(E_0 + \frac{2P}{R^3} \right) = 4\pi\sigma(\theta)$$

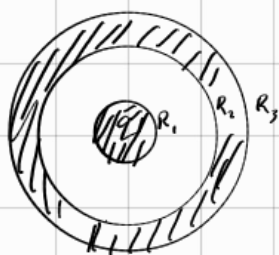
$$E_{\tau out} = E_0 \sin\theta - \frac{P \sin\theta}{R^3} = 0 \quad P = E_0 R^3$$

$$4\pi\sigma(\theta) = \cos\theta \left(E_0 + \frac{2}{R^3} E_0 R^3 \right) = 3 E_0 \cos\theta$$

$$\sigma = \frac{3}{4\pi} E_0 \cos\theta$$

$$E_{out} = \vec{E}_0 + \frac{3(E_0 R^3 \vec{r})\vec{r}}{R^5} - \frac{E_0 R^3 \vec{r}}{R^3} = 3 E_0 \cos\theta \quad \left\{ \begin{array}{l} E_0 \neq 3 \cos\theta = 1 \\ 2 E_0 \neq 3 \cos\theta = 2 \end{array} \right.$$

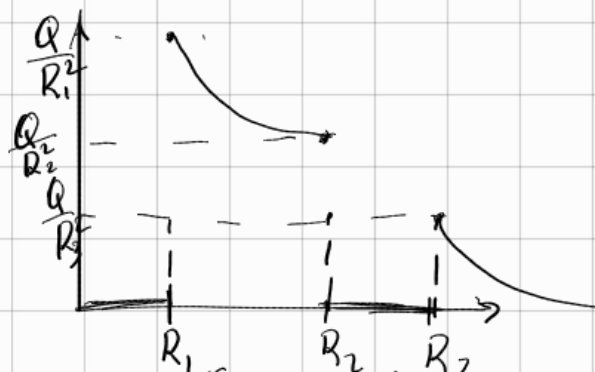
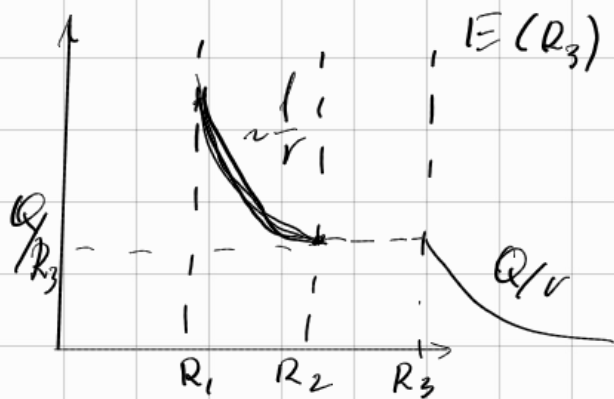
2.3



$$\oint E_n dS = 4\pi Q$$

$$E(R_3) \cdot 4\pi R^2 = 4\pi Q$$

$$E(R_3) = \frac{Q}{R_3^2}$$



$$\phi = -\int E dr = -\int \frac{Q}{r^2} dr = \frac{Q}{r}$$

$$\phi(r) = \frac{Q}{r} + C$$

$$\phi(R_2) = \frac{Q}{R_2} + C = \frac{Q}{R_3} \quad C = Q \left(\frac{1}{R_3} - \frac{1}{R_2} \right)$$

